Prodigy Anesthesia Quick Review

von Willebrand Disease

What is von Willebrand disease, how common is it, and does it affect males or females more often?	von Willebrand disease is an inherited platelet function disorder caused by a deficiency in von Willebrand factor that affects males and females equally and occurs in about 1% of the population *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 209.*
What is the function of von Willebrand factor?	It serves as a carrier for factor VIII and acts to protect it from proteolytic cleavage. It also binds platelet receptors to exposed collagen initially by a Ib-vWF and collagen interaction and eventually assists in a stronger binding between platelets and collagen via the IIb-IIIa-vWF and collagen interaction. It also acts as a bridge between the glycoprotein IIb/IIIa receptors to promote platelet aggregation. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 209.*
What are the signs and symptoms of von Willebrand disease?	There are three major forms of the disease that range from mild to severe, so symptoms may not appear as abruptly or severely in all patients. However, typical symptoms include easy bruising, nose bleeds, gastrointestinal bleeding, menorrhagia, and excessive bleeding during surgical procedures. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 209.*
How is von Willebrand disease treated?	The most common treatment is desmopressin 0.3 mcg/kg IV in 50 mL of saline over 20 minutes. It may also be administered as a nasal spray. Other treatment options include vWF concentrates, alpha-aminocaproic acid, and tranexamic acid. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 209-210.*
Besides desmopressin and vWF concentrate, what is another treatment option for a patient suffering from von Willebrand disease?	Cryoprecipitate may be administered as it contains concentrated fibrinogen, fibronectin, vWF, and factors VIII and XIII. It should be noted, however, that commercially prepared concentrates of factor VIII-vWF are preferred as cryoprecipitate still carries a risk of transfusion-transmitted infection. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 429.*
What side effects of DDAVP administration in a patient with von Willebrand disease could you expect to see?	Too rapid administration of DDAVP IV can cause lightheadedness, flushing, headache, and nausea. It also possesses a mild antidiuretic effect and can result in water intoxication in patients receiving multiple DDAVP treatments or large volumes of fluids. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 429.*

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What drugs should be avoided in the patient with von Willebrand disease?	No specific anesthetic agents are contraindicated in these patients, however, it should be noted that the administration of NSAIDs such as ketorolac can adversely affect an already impaired platelet function and result in excessive or spontaneous bleeding. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 209-210.*
How are coagulation assays affected in patients with von Willebrand disease?	They will display a low von Willebrand factor concentration (the normal is 5-10 mg/dL), a prolonged bleeding time, prolonged PTT, and a decreased factor VIII activity. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 428.*
Besides the hereditary form of von Willebrand disease, what other ways might it appear?	Acquired von Willebrand disease can occur in patients with some tumors that absorb von Willebrand factor into the tumor surface. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 705.*
What anesthetic implications should you consider in the pregnant patient with von Willebrand disease?	When considering anesthetic type, it is worth considering the increased risk for a spinal or epidural hematoma when performing neuraxial anesthesia. Normally, vWF factors increase two or three-fold during the second and third trimester, which afford a degree of protection from the risk of bleeding, but this should be verified with laboratory testing if possible. Also, intranasal desmopressin is safe for use in pregnant patients. Some individuals, however, have a poor response to DDAVP and should be treated with vWF concentrates. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 210.*

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